Exp No: 18

GENERATION OF PWM USING TIMER INTERRUPT

AIM:

To generate PWM using Timer interrupt.

APPARATUS / SOFTWARE REQUIRED:

• PC with Windows OS

• Keil µVision IDE

• 8051 Simulator (Keil built-in)

• Serial Window (UART #1)

PROGRAM:

```
#include &lt;reg51.h&gt;

sbit PWM = P1^0;

unsigned char count = 0;
unsigned char duty = 100; // Change this value (0–200)

void timer0_ISR(void) interrupt 1
{
TH0 = 0xFC; // Reload for 1ms delay
TL0 = 0x66;

count++;

if(count &lt; duty)
PWM = 1; // ON time
else
```

```
PWM = 0; // OFF time

if(count &gt;= 200) // Total period = 200ms
count = 0;
}

void main()
{
TMOD = 0x01; // Timer0 Mode 1 (16-bit)
TH0 = 0xFC; // 1ms preload
TL0 = 0x66;

IE = 0x82; // EA = 1, ET0 = 1
TR0 = 1; // Start Timer0

while(1);
}
```

Output